



Igor Santana

MSc in Computer Science · Senior Software Engineer

igorpsant95@gmail.com
Maringá, Brazil

// SKILLS

FRONTEND

- TypeScript
- JavaScript
- Angular
- React
- Vue.js
- Ionic

ARCHITECTURE

- Micro frontends
- Design systems
- REST APIs

TOOLING

- Node.js
- Docker
- AWS
- GitHub Actions
- CI/CD

BACKGROUND

- Python
- Java
- Machine learning

LANGUAGES

- Portuguese (native)
- English (professional)

// SUMMARY

Senior software engineer with 10+ years of frontend experience across Angular, React, and Vue.js, in varied settings: mobile applications in agribusiness, financial products, and international enterprise platforms in logistics.

I have led design system and micro frontend initiatives, focusing on scalable UI architecture and shared standards that help teams ship reliably at scale.

I hold an MSc in computer science with published research on recommender systems, *deep learning*, contextual embeddings, and neural network models for personalized recommendation. This work is closely related to the foundations behind today's AI and LLM-driven products.

// PROFESSIONAL EXPERIENCE

- Senior Software Engineer · Remote

Zup Innovation

Feb 2025 – Present

Allocated to Itaú Unibanco's real estate collateral squad, I develop micro frontends with Angular in cross-functional teams. On my previous squad, I worked in the agricultural sector, also building micro frontends.
- Senior Software Engineer · Lisbon, Portugal (Remote)

GLS Portugal

Mar 2022 – Nov 2024

I helped define and build a design system to accelerate delivery across European branches, contributed to the CORE Platform for micro frontends, and worked with international teams to adopt shared components and standards.
- Mid-level Software Engineer · Remote

Objective

Nov 2021 – Feb 2022

I developed frontend applications with React and Liferay for an international project in the health insurance area. I built user-facing features for portal and web experiences, working with distributed teams across countries.

// ACADEMIC EXPERIENCE

- MSc in Computer Science

Universidade Estadual de Maringá

2018 – 2020

Master's thesis exploring recurrent neural networks to generate contextual embeddings from users' listening sequences. The embeddings were applied to four context-aware music recommender systems and evaluated on two datasets, outperforming the baseline model across all metrics used.
- BSc in Informatics

Universidade Estadual de Maringá

2013 – 2017

Undergraduate thesis proposing data mining and clustering techniques to automatically extract contextual information from georeferenced data. The work compared clustering algorithms and evaluated their impact on context-aware recommender systems using Foursquare and Yelp datasets.

Mid-level Software Engineer · Remote
Accountfy

I owned the frontend of the Cash Management module, building financial systems with Angular in a fintech product. I delivered features for cash flow management and banking operations, collaborating with product, design, and backend teams.

Mid-level Software Engineer · Maringá, Brazil
FarmGO Agrosolutions

I developed mobile applications with Ionic and Vue.js for field use in agribusiness, and refactored legacy frontend and backend systems built with Django and jQuery.

Entry-level Software Engineer · Maringá, Brazil
Gumga Tecnologia

I worked on a web framework with Angular 1.x and Bootstrap adopted by partner companies, delivering reusable UI components and shared patterns. I also contributed to a code generator that accelerated new project setup and reduced boilerplate across client engagements.

Undergraduate Researcher · Maringá, Brazil
CNPq

Expanded abstract on combining the Combined Reduction and DaVI context-aware recommendation algorithms to improve prediction quality over each method alone. Implemented in Java using the CARSKit framework.

// PUBLICATIONS

Music4All: A New Music Database and Its Applications

27th International Conference on Systems, Signals and Image Processing (IWSSIP), 2020 · 121 citations

Introduces Music4All, a music database with metadata and contextual listening data, and demonstrates its applications in music recommendation research.

A Systematic Review on Context-Aware Recommender Systems Using Deep Learning and Embeddings

arXiv:2007.04782, 2020 · 15 citations

Systematic review of context-aware recommender systems that use deep learning and embeddings, mapping methods, datasets, and open challenges in the field.

Context-Aware Music Recommendation with Metadata Awareness and Recurrent Neural Networks

Computing and Informatics 41 (3), 834-860, 2022 · 8 citations

Proposes Metadata-Aware and recurrent neural network methods to extract contextual embeddings from listening sequences, evaluated across four context-aware music recommenders.

Transforming Geo-Referenced Data in Contextual Information for Context-Aware Recommender Systems

IEEE/WIC/ACM International Conference on Web Intelligence (WI), 2018 · 1 citation

Applies clustering to transform georeferenced data into contextual regions for context-aware recommender systems, evaluated on Foursquare and Yelp datasets.

Improving Context-Aware Music Recommender
Systems with a Dual Recurrent Neural Network
Information Management and Big Data – SIMBig
2020, 2021

Uses a dual recurrent neural network to
generate contextual embeddings from listening
sequences, improving four context-aware music
recommenders on two datasets.